

RED CELLS, IRON, AND ERYTHROPOIESIS

Direct and indirect regulation of fetal globin transcript by the RNA-binding protein IGF2BP1

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KEY POINTS

- The RBP regulates fetal hemoglobin directly through posttranscriptional control and indirectly via BCL11A.
- IGF2BP1 binding to a stop codon–proximal site in the coding sequence of the *HBG1/2* globin transcript supports fetal hemoglobin translation.

Despite extensive investigation, the molecular control of developmental hemoglobin expression remains incompletely elucidated. Hemoglobin switching is controlled by transcription factors, microRNAs, and RNA-binding proteins (RBPs) that enforce gene regulatory changes through development. Here, we examine the role of the heterochronically silenced N6 methyladenosine (m6A) RBP insulin-like growth factor 2 mRNA-binding protein 1 (IGF2BP1) that was previously described to regulate *HBG1/2* indirectly by suppressing *B-cell lymphoma 11A* (*BCL11A*) expression through an unknown mechanism. We find that IGF2BP1 binds and activates hypermethylated in cancer 2 (*HIC2*), itself a *BCL11A* repressor. Furthermore, we identify that IGF2BP1 plays a *BCL11A*-independent role by direct binding to *HBG1/2* to promote its translation. Stop codon–proximal m6A-modified coding sequences within *HBG2* transcripts are necessary and sufficient for direct positive regulation mediated by IGF2BP1. This work deepens the mechanistic understanding of hemoglobin switching and suggests a physical relationship between heterochronic RBPs and globin transcripts.

Introduction

The developmental timing of globin gene expression and, of particular clinical interest, the switch from fetal hemoglobin (HbF) to adult hemoglobin, is a coordinately controlled process, whereby the fetal *HBG1/2* genes are silenced and adult *HBB* gene is reciprocally activated around birth.¹⁻³ This transition is mediated by genes that are themselves heterochronically regulated and that support not just the fetal-to-adult hemoglobin switch but developmentally regulated gene expression programs in many cell contexts.⁴⁻¹⁰ These heterochronic factors function in a network regulating, and regulated by, insulin-like growth factor 2 mRNA-binding protein 1 (IGF2BP1) and include the transcription factor hypermethylated in cancer 2 (*HIC2*) (directly regulating *BCL11A* transcription), *HMGA2* (unclear mechanism), *LIN28B* (regulating *BCL11A* translation), and adult-stage *let-7* microRNA (miRNA; regulating *HIC2*, *LIN28B*, *IGF2BP1*, and *HMGA2*).^{4-8,10,11} Notably, the miRNA *let-7* targets all of these heterochronic HbF regulators including *IGF2BP1*.^{5,7,11-14}

The N6 methyladenosine (m6A) RNA-binding protein (RBP) *IGF2BP1* is developmentally silenced after the fetal stage across many tissues, including in erythroid precursors during

the hemoglobin switch.^{8,15} m6A modifications are written on RNAs by the *METTL3/METTL14* heteroduplex at DRACH motifs.¹⁶ Most m6A that is deposited on transcripts is observed in coding sequences (CDS) near the stop codon and in the 3' untranslated region (UTR). Study of regulatory m6A has been focused on the 3' UTR.¹⁷⁻¹⁹ Recent work has identified CDS–m6A decay, in which the m6A RBP *YTHDF2* binds to stop codon–proximal m6A in the coding region to degrade transcripts.²⁰ Before the appreciation of m6A as a dynamic regulatory mark and the identification of *IGF2BP1* as an m6A RBP, the coding region determinant of instability (CRD) in cellular myelocytomatosis oncogene (*c-MYC*) mRNA was characterized as a target of *IGF2BP1* (initially known as CRD-BP), which binds and stabilizes the otherwise unstable *c-MYC* transcript, suggesting that *IGF2BP1* in addition to *YTHDF2* can contribute to control of gene expression through CDS m6A.²¹⁻²⁴

BCL11A is a key orchestrator of hemoglobin switching, binding the *HBG1/2* promoters and repressing *HBG1/2* transcription.²⁵⁻²⁸ Mutations in the binding sites of *BCL11A* at the –115 region of the *HBG1/2* promoters have been found to be causative of hereditary persistence of HbF.²⁸ *BCL11A* recruits the NuRD complex to the *HBG1/2* gene loci to enable globin switching.²⁹ An erythroid-specific enhancer controlling

BCL11A gene expression is the target of the first US Food and Drug Administration–approved CRISPR gene therapy for transfusion-dependent β -thalassemia and sickle cell disease.^{30–33}

Although m6A was first identified as a posttranscriptional RNA modification in 1972, the identification of FTO (fat mass and obesity–associated protein) as an m6A demethylase decades later led to the understanding that m6A is a dynamically regulated mark with a diverse set of roles involved in post-transcriptional control of gene expression.^{34–36} The m6A writing complex is a heteroduplex composed of METTL3 and METTL14, aided by several cofactors.³⁶ FTO and AlkB homolog 5 (ALKBH5) have been identified as m6A demethylases.³⁶ In addition to writers and erasers of m6A, readers include IGF2BP1/2/3, YTHDF1/2/3, YTHDC1/2, H2RNPC, and HNRNPA2B1.^{19,36–38} Binding of IGF2BP family RBPs is generally associated with increased translation and/or RNA stability, whereas binding of YTHDF family proteins is generally associated with destabilization and RNA decay; however, the gene regulatory outcome for individual transcripts or contexts may not be predictable or completely understood.^{39,40} Notably, loss of expression of the *METTL3* gene component of the m6A writer complex leads to hematopoietic defects in both zebrafish and mice.²³

Mechanistically, IGF2BP1 supports the expression of bound mRNAs by increasing their mRNA stability and/or translation efficiency.³⁹ In erythroid cells, however, *IGF2BP1* overexpression has been shown to have a role in depressing the expression of the key globin switching transcription factor *BCL11A* in adult erythroid cells. IGF2BP1 was shown to bind to *BCL11A* mRNA yet *BCL11A* mRNA stability and translation were unchanged, suggesting an uncertain mechanism.^{8,30,41,42} Also, in this previous report, IGF2BP1 overexpression resulted in HbF levels up to 90% of total hemoglobin, notably higher than previously demonstrated upon loss of *BCL11A* expression.^{8,43} Both the downregulation of *BCL11A*, opposite to the typical outcome of *IGF2BP1* binding to a target mRNA, as well as the magnitude of the HbF upregulation upon *IGF2BP1* overexpression raise the question of additional mechanisms besides loss of *BCL11A* expression.

Methods

See the supplemental Methods, available on the *Blood* website, for additional detail.

IGF2BP1 cross-linking and immunoprecipitation (CLIP) followed by RT-qPCR (CLIP-qPCR)

Suspension cells (2e7) were collected, cross-linked at 4000 $\mu\text{J}/\text{cm}^2$, and flash frozen. Lysates were split for matched pulldown with IGF2BP1 or control antibody. After alternating high and low salt washing, RNA was liberated from beads, column purified, and measured by reverse transcription and quantitative polymerase chain reaction (RT-qPCR).

Polysome profiling

Human umbilical–derived erythroid progenitor (HUDEP) with an adult erythroid gene expression profile (HUDEP-2) cells transduced with *PLVX-IGF2BP1-Puro* or empty vector, selected with puromycin (10 $\mu\text{g}/\text{mL}$), then were treated with

cycloheximide (100 $\mu\text{g}/\text{mL}$) to arrest translation. Lysate was loaded onto 10% to 50% sucrose gradients and ultracentrifuged before fractionation. Firefly luciferase mRNA prepared by in vitro transcription was spiked into each fraction before RNA purification, reverse transcription, and RT-PCR.

HiBiT luminescence assays

The *HiBiT-HBG2* and *HiBiT-HBB* constructs were cloned using synthetic gene fragments containing respective 5' and 3' UTRs and a 5' HiBiT tag (Promega), joined by Gibson assembly.⁴⁴ HUDEP or CD34⁺ hematopoietic stem and progenitor cells (HSPCs) were transduced with HiBiT-globin lentivirus and analyzed using the Nano-Glo HiBiT Lytic Detection System (Promega).

RNA sequencing (RNA-seq)

HUDEP-1 and HUDEP-2 cells transduced with either *PLVX-SFFV-IGF2BP1-Puro* or empty vector were differentiated in erythroid differentiation media II for 8 days before TruSeq library preparation and paired-end 150–base pair sequencing. Differential gene expression analysis was performed using DESeq2 (National Center for Biotechnology Information GSE303945).^{45,46}

m6A MeRIP

m6A methylated RNA immunoprecipitation (MeRIP) was performed as previously described.⁴⁷ Poly-A⁺ mRNA was mixed with anti-m6A antibody or normal rabbit immunoglobulin G and then protein A beads to retrieve m6A-bound mRNA. Beads were washed before digestion of protein by proteinase K. After proteinase K treatment and reverse transcription, samples were analyzed by RT-PCR.

Isolation of cord blood reticulocyte RNA and direct RNA-seq

Cord blood units were obtained with institutional review board approval, and reticulocytes isolated.⁴⁸ Control *HBG2* mRNA containing 0% or 100% m6A was generated by in vitro transcription with N6-methyladenosine 5' triphosphate (TriLink Biotech).

Direct RNA-seq used Oxford Nanopore Technologies sequencing. m6A basecalling used the Dorado (version 0.9.1) *ma004_130bps_sup@v5.1.0_m6A_DRACH@v1* model.⁴⁹

Results

BCL11A-independent regulation of HbF by IGF2BP1

To investigate the role of IGF2BP1 in adult-stage globin expression, we overexpressed *IGF2BP1* in immortalized HUDEP-2 cells.⁵⁰ We observed an increase in *HBG1/2* mRNA level, consistent with previous findings (Figure 1A). To test whether *BCL11A* was required for this upregulation in HbF expression, we overexpressed IGF2BP1 in *BCL11A*-knockout (Δ *BCL11A*) HUDEP-2 cells. These cells have increased baseline *HBG1/2* mRNA and HbF levels as expected. Nonetheless, upon IGF2BP1 overexpression, we observed a further increase in *HBG1/2* mRNA and of HbF levels as compared with empty vector control cells (Figure 1B–C). This suggests a role for *IGF2BP1* in regulating HbF expression that is

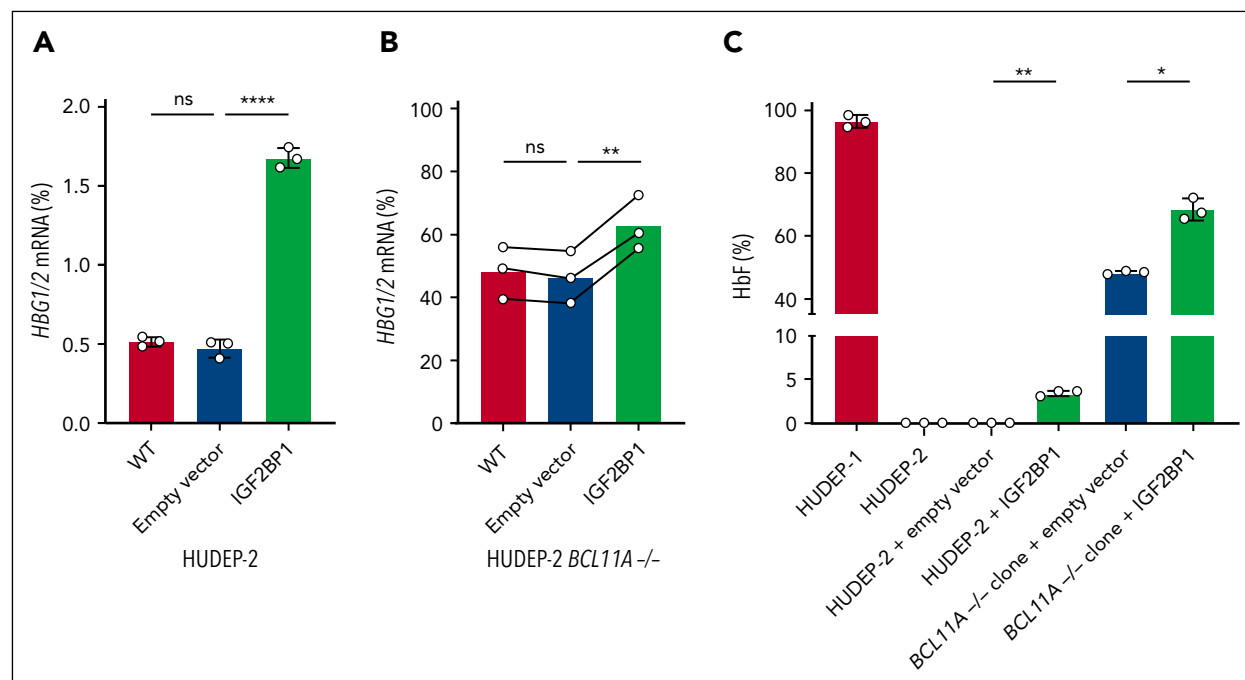


Figure 1. BCL11A-independent regulation of HbF by IGF2BP1. (A) Expansion phase HUDEP-2 cells were transduced with control or IGF2BP1 expression vector using lentivirus. Primers targeting *HBG1/2* and *HBB* were used to measure the RNA level. *HBG1/2* mRNA level is reported as a percent of total *HBB* + *HBG1/2* mRNA. (B) Expansion phase HUDEP-2 *BCL11A*^{-/-} clones were transduced and RT-qPCR was performed as in panel A. (C) Indicated cell lines were erythroid differentiated in vitro for 12 days and hemoglobin high-performance liquid chromatography was performed to quantify the level of HbF expressed as percentage of HbF + HbA. All *P* values were calculated using 1-way ANOVA. **P* < .05; ***P* < .01; *****P* < .0001. ns, not significant.

BCL11A-independent in addition to the previously described regulation of *BCL11A* by *IGF2BP1*.

IGF2BP1 overexpression indirectly regulates *BCL11A* by supporting *HIC2* expression

m6A is a dynamically regulated RNA modification for which most described regulatory sites for protein coding transcripts are found within the 3' UTR, and most observed m6A modifications lie within the 3' UTR or in proximity to the stop codon in the last coding exon.^{18,35,36,51-53} The regulatory outcome of any given m6A mark is primarily driven by the identity of the m6A RBP that binds the mark.^{36,37} *IGF2BP1* is an m6A RBP, which supports the RNA stability and/or the translation efficiency of bound targets.^{14,19,39,54,55} *IGF2BP1* overexpression suppresses *BCL11A* expression, which raises the possibility that *IGF2BP1* may indirectly control *BCL11A* expression.⁸

Recent work has identified a role for the transcription factor *HIC2* in repression of *BCL11A* transcription, and *HIC2* is itself silenced by the heterochronic adult-stage miRNA *let-7*, which is also known to negatively regulate embryonic/fetal-stage factors *LIN28B* and *IGF2BP1*.^{4,12,56} We speculated that *HIC2* may be the direct target of *IGF2BP1* responsible for the *IGF2BP1*-mediated suppression of *BCL11A* upon *IGF2BP1* overexpression in adult erythroblasts. *IGF2BP1* overexpression in HUDEP-2 cells in vitro differentiated for 4 days led to an upregulation of *HIC2* protein as well as a reduction in *BCL11A* protein (Figure 2A-B). To test whether the reduction in *BCL11A* expression upon *IGF2BP1* overexpression required *HIC2*, we generated *HIC2*^{-/-} HUDEP-2 clones. Overexpression of

IGF2BP1 in these cells failed to reduce *BCL11A* expression, suggesting that *HIC2* is required for the effect of *IGF2BP1* overexpression on *BCL11A* (Figure 2C-D). Consistent with that mechanism, overexpression of *IGF2BP1* in HUDEP-2 cells reduced *BCL11A* mRNA level (supplemental Figures 1B-C and 2A-B). Overexpression of *IGF2BP1* in HUDEP-2 cells led to increased *HIC2* expression as measured by RT-qPCR (Figure 2E). To determine whether *IGF2BP1* binds directly to *HIC2* mRNA, we performed *IGF2BP1* CLIP-qPCR on HUDEP-2 cells overexpressing *IGF2BP1*. We observed enrichment of *HIC2* mRNA bound to *IGF2BP1* (Figure 2F-G). To test whether the effect of *IGF2BP1* overexpression may increase HbF expression in the *HIC2*^{-/-} setting, we performed flow cytometry on *HIC2*^{-/-} cells expressing *HBG2-2A-mCherry* after transduction with control empty vector or *IGF2BP1* overexpression vector (Figure 2H). We observed an increase of mCherry fluorescence after *IGF2BP1* overexpression in *HIC2*^{-/-} cells, consistent with a *BCL11A*-independent component of the effects of *IGF2BP1* on HbF.

IGF2BP1 directly regulates *HBG1/2* mRNA posttranscriptionally

To identify candidate *IGF2BP1* target genes responsible for the *BCL11A*-independent regulation of HbF, we performed differential gene expression analysis upon overexpression of *IGF2BP1* in HUDEP-2 cells. Additionally, we compared these data sets with RNA-seq data from primary human fetal liver, cord blood, adult bone marrow, as well as wild-type (WT) HUDEP-1 and HUDEP-2 cells. Overexpression of *IGF2BP1* resulted in a shift of HUDEP-2 cells toward a fetal-like

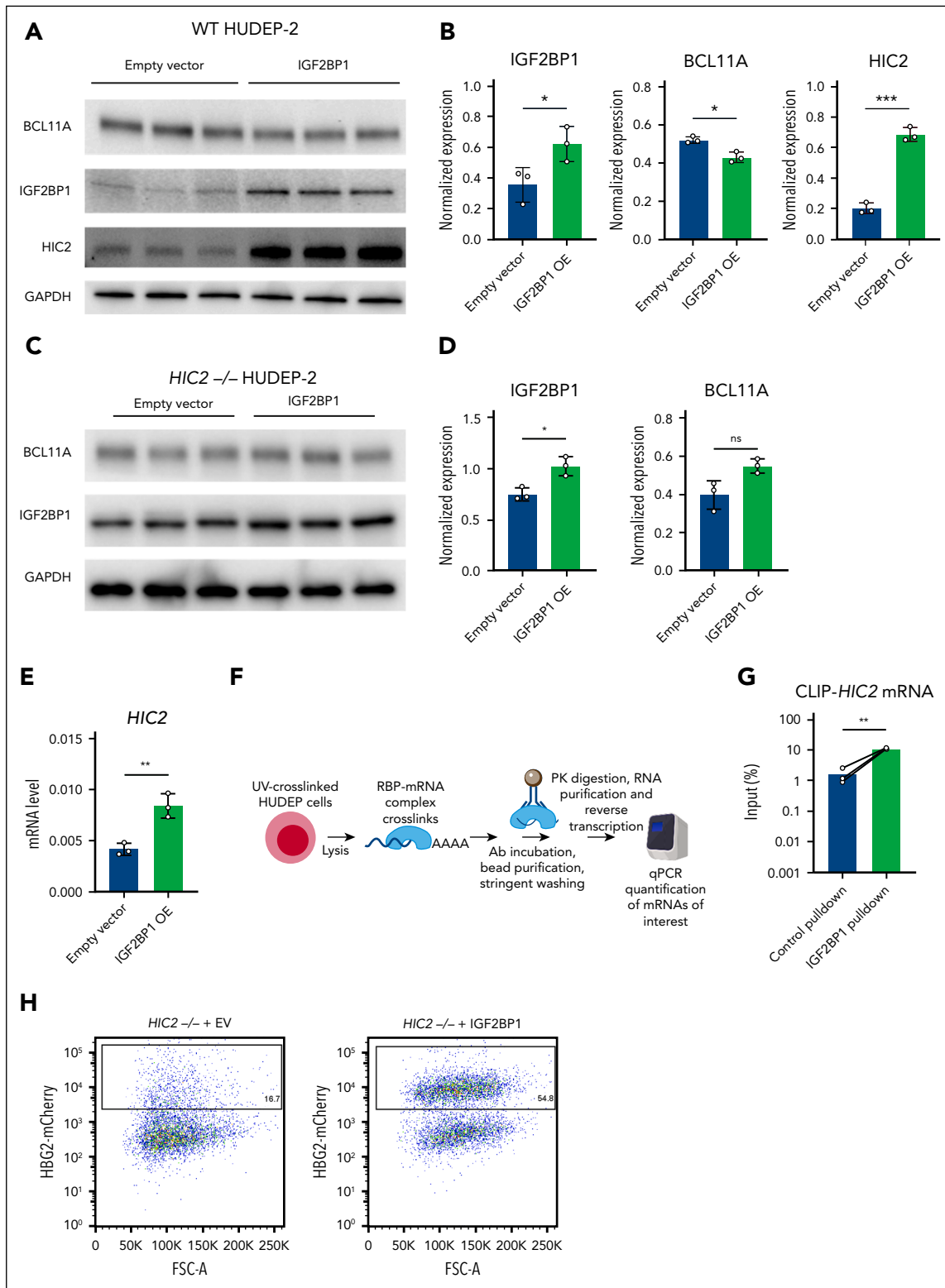
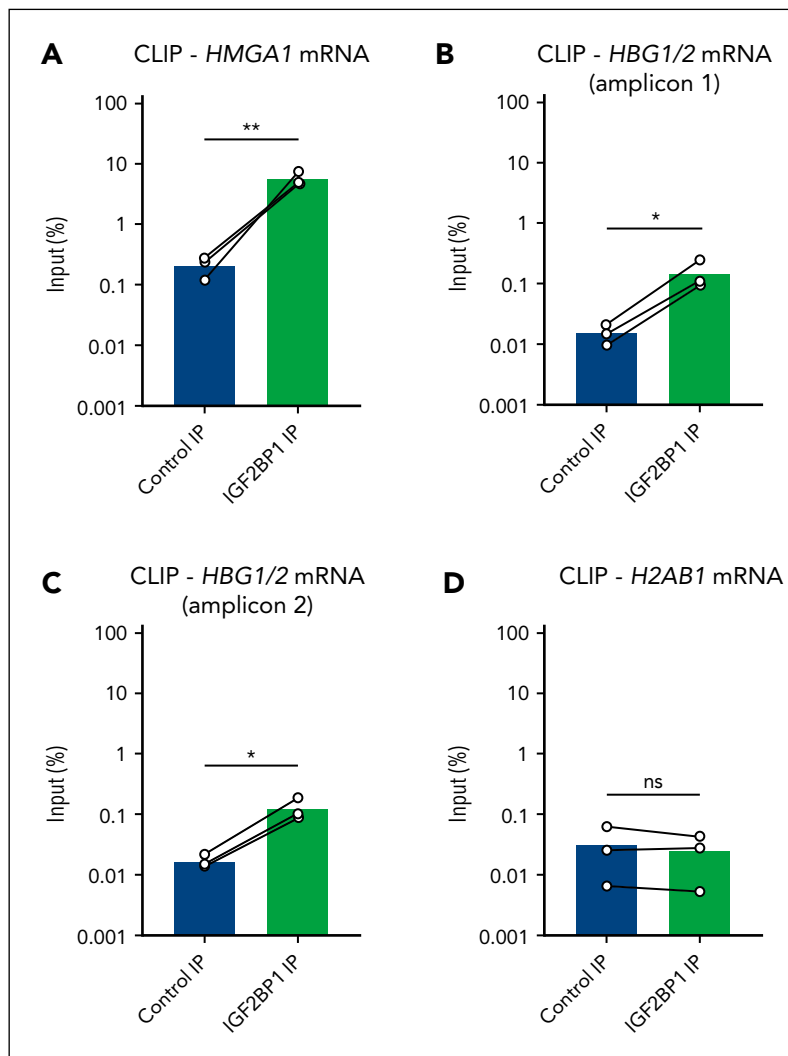


Figure 2. The BCL11A-dependent effect of IGF2BP1 on HbF regulation is mediated by HIC2. (A) HUDEP-2 cells were transduced in triplicate with control or IGF2BP1 overexpression vectors and differentiated in vitro for 4 days before protein was isolated for western blotting of indicated proteins. (B) Densitometry of indicated proteins normalized to GAPDH expression from panel A. (C) HbG2-2A-mCherry HIC2 $-/-$ HUDEP-2 cells were transduced in triplicate with control or IGF2BP1 overexpression and protein was isolated for western blotting of indicated proteins. (D) Densitometry of indicated proteins normalized to GAPDH expression from panel B. (E) HIC2 mRNA level from HUDEP-2 cells transduced with control or IGF2BP1 overexpression vector was measured by RT-PCR relative to catalase. (F) Schematic of the IGF2BP1 CLIP-qPCR experiment with expansion phase HUDEP-2 cells. (G) IGF2BP1 CLIP-qPCR was performed from HUDEP-2 cells and HIC2 mRNA level was measured. (H) HbG2-2A-mCherry HIC2 $-/-$ HUDEP-2 cells were treated with control empty vector or IGF2BP1 overexpression vector and mCherry-high cells were measured by flow cytometry. * $P < .05$; ** $P < .01$; *** $P < .001$. GAPDH, glyceraldehyde-3-phosphate dehydrogenase; ns, not significant; OE, overexpression.

Figure 3. IGF2BP1 CLIP-qPCR. (A) IGF2BP1 CLIP-qPCR performed in HUDEP-1 cells measuring known *IGF2BP1* target *HMGA1*. (B) *HBG1/2* spanning the exon 2/exon 3 junction. (C) *HBG1/2* using primers targeting exon 3. (D) *H2AB1*. * $P < .05$; ** $P < .01$. IP, immunoprecipitation; ns, not significant.



expression profile primarily driven by increases in the expression of developmentally regulated hemoglobin gene expression, specifically *HBZ*, *HBE1*, *HBG1/2*, and the *HBBP1* pseudogene (supplemental Figure 1). Upregulation of embryonic globins *HBZ* and *HBE1* upon *IGF2BP1* overexpression was confirmed by RT-qPCR (supplemental Figure 2), consistent with a reduced level of *BCL11A* expression.^{43,57,58} Profiling of known HbF regulators by RNA-seq failed to reveal obvious candidates responsible for the *BCL11A*-independent regulation. Given its function as a positively regulating RBP, we hypothesized that *IGF2BP1* might directly bind to *HBG1/2*. We performed *IGF2BP1* CLIP-qPCR from HUDEP-1 cells, HUDEP cells with a fetal-like expression pattern (Figure 3).⁵⁰ We observed significant binding of *IGF2BP1* to its known target *HMGA1* (Figure 3A). We also found significant binding of *IGF2BP1* to the *HBG1/2* mRNA when measured using primers for an amplicon spanning the exon 2/exon 3 junction of *HBG1/2* (Figure 3B) as compared with the negative control histone mRNA *H2AB1* (Figure 3D). A separate pair of primers targeting the *HBG1/2* exon 3 CDS yielded a similar result in terms of *IGF2BP1* binding (Figure 3C). These results raised the possibility of a direct role for *IGF2BP1* in regulating *HBG1/2* posttranscriptionally.

***IGF2BP1* regulates *HBG1/2* mRNA directly through a stop codon-proximal DRACH motif**

To distinguish the indirect *BCL11A*-dependent effect of *IGF2BP1* on *HBG1/2* transcription from the possible direct regulatory effects of *IGF2BP1* binding to the *HBG1/2* mRNA, we designed a lentiviral reporter system independent of endogenous globin gene transcription. This *SFFV-HiBiT-HBG-PGK-enhanced green fluorescent protein (eGFP)* reporter construct uses a HiBiT tag at the 5' end of the *HBG2* protein coding region which enables transgene-specific mRNA and protein level quantification. To test whether *IGF2BP1* can upregulate *HiBiT-HBG* expression, we transduced HUDEP-2 cells with the *HiBiT-HBG* reporter vector (Figure 4A). Then the culture was split before subsequent transduction with *IGF2BP1* or control vector. Upon *IGF2BP1* overexpression, we observed an increase in luminescence as compared with control cells (Figure 4B). To test whether this effect was *BCL11A* independent, we transduced *BCL11A*^{-/-} HUDEP-2 clones. We observed a similar increase in HiBiT-HBG luminescence upon *IGF2BP1* overexpression in *BCL11A*^{-/-} HUDEP-2 clones, demonstrating that this regulation is *BCL11A*-independent (Figure 4C). This effect size was similar to the experiment in

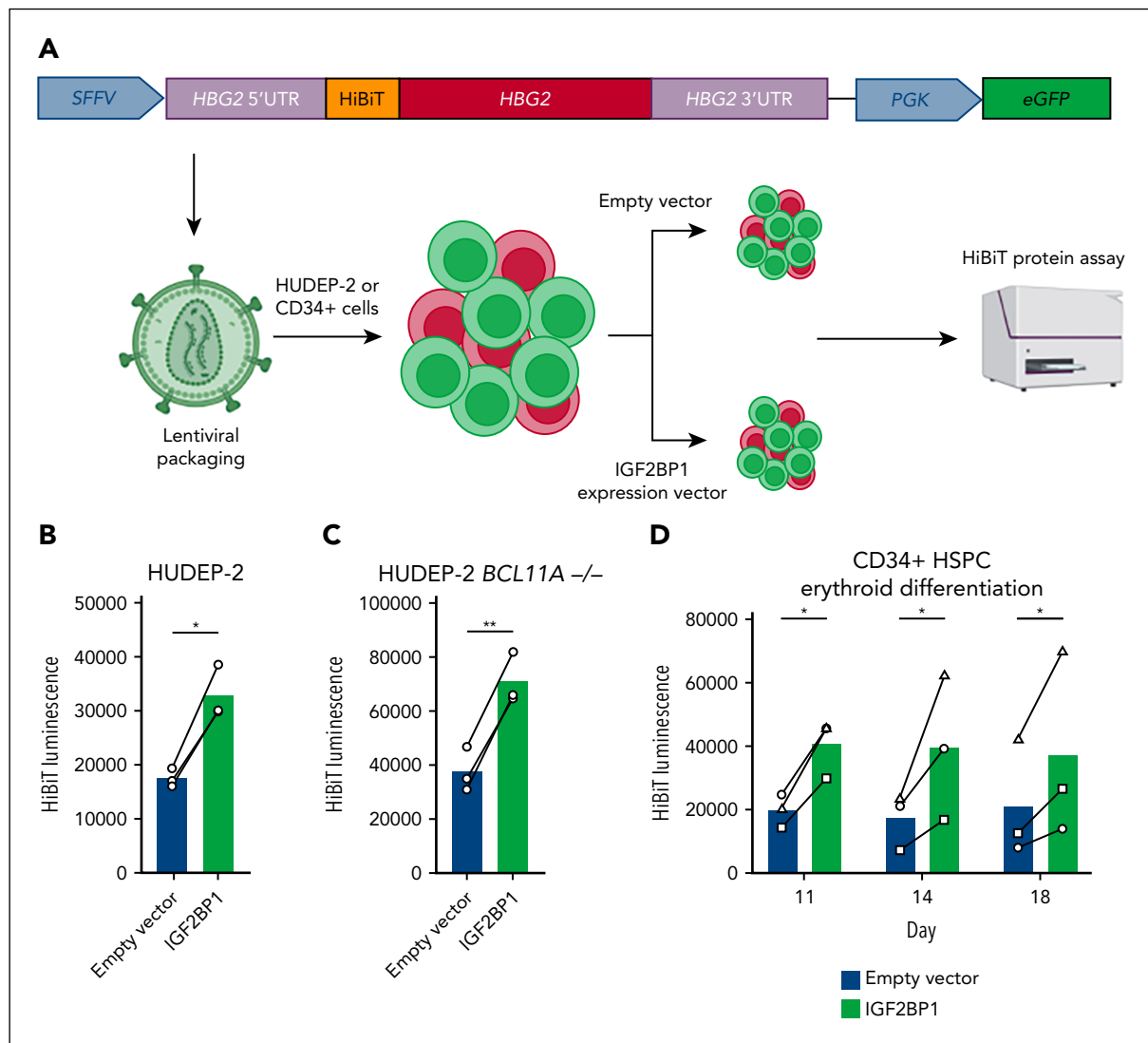


Figure 4. IGF2BP1 regulates HBG1/2 level independent of endogenous transcriptional regulation. (A) Lentivirus containing the SFFV-HiBiT-HBG-PGK-eGFP cassette was used to transduce HUDEP-2 or CD34⁺ HSPCs. (B-D) Each transduction was split for a second round of transduction with IGF2BP1 or empty control vector, then luminescence was measured after selection for WT HUDEP-2 cells (B), Δ BCL11A HUDEP-2 cells (C), and in vitro erythroid differentiated CD34⁺ HSPCs (D) measured at the indicated time points. * $P < .05$; ** $P < .01$. eGFP, enhanced green fluorescent protein; ns, not significant; PGK, phosphoglycerate kinase promoter; SFFV, spleen focus-forming virus promoter.

BCL11A WT cells. Next, we tested whether this regulation was intact in primary CD34⁺ HSPCs. We transduced CD34⁺ HSPCs from 3 donors with the *HiBiT-HBG* reporter vector and control or *IGF2BP1* overexpression vector and observed a similar increase in *HiBiT-HBG* luminescence upon *IGF2BP1* overexpression throughout in vitro erythroid differentiation (Figure 4D). These results suggest that *HBG1/2* regulation by *IGF2BP1* is independent from the endogenous transcriptional regulation by *BCL11A* or other transcriptional regulators of the endogenous β -globin gene cluster.

IGF2BP1 is an m6A-RBP that acts to stabilize its targets and promote their expression by improving translation efficiency.^{19,21,39,59,60} Given the observations that *IGF2BP1* directly binds to, and promotes the expression of, *HBG1/2* mRNA, we performed polysome profiling to test whether this *BCL11A*-independent mechanism acted by promoting translation. HUDEP-2 cells were transduced with SFFV-*HiBiT-HBG2*-PGK-eGFP reporter lentivirus (*HiBiT-HBG*) before further

transduction with *IGF2BP1* or empty vector. Lysates from these cells were subjected to polysome profiling and were fractionated into 24 fractions (Figure 5B). After fractionation, we isolated RNA and measured the abundance of *HiBiT-HBG* reporter mRNA in each fraction. We grouped fractions corresponding to the following 3 groups: polysome free, early polysome, and late polysome. We observed a significant increase in the abundance of reporter mRNA in late-polysome fractions upon *IGF2BP1* overexpression consistent with increased translation of the transgenic *HiBiT-HBG* mRNA mediated by *IGF2BP1* (Figure 5).

Erythroid maturation is characterized by substantial monotonic increase of hemoglobin level. Therefore, changes in erythroid differentiation kinetics resulting from *IGF2BP1* overexpression might confound interpretation of specific changes in globin gene expression. We tested whether *IGF2BP1* overexpression had an impact on in vitro erythroid differentiation of CD34⁺ HSPCs and observed no difference in the CD71⁺CD235a⁺

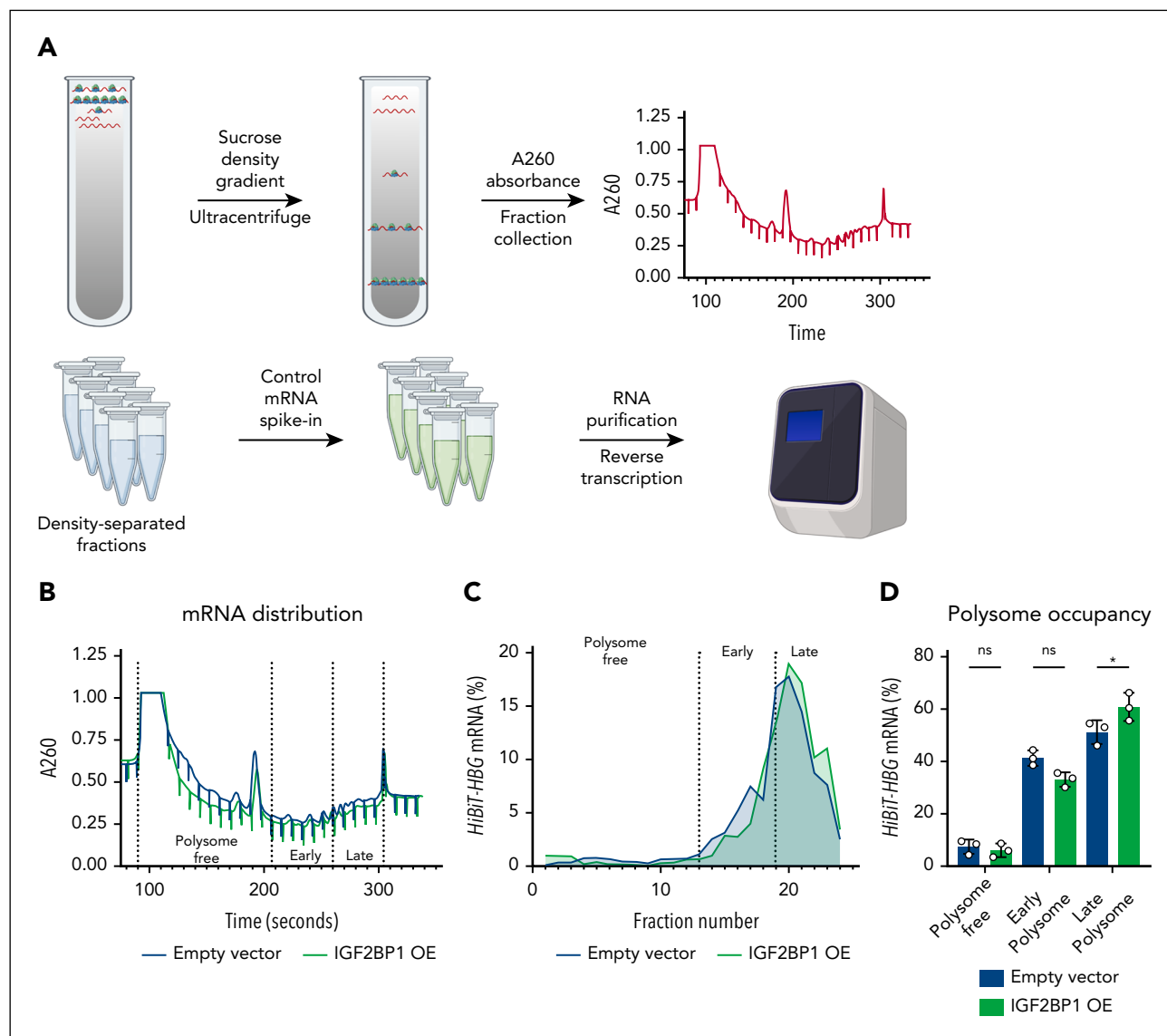


Figure 5. IGF2BP1 promotes translation of *HBG1/2* mRNA. Expansion phase HUDEP-2 cells transduced with HiBiT-HBG reporter virus were split and further transduced with IGF2BP1 or control overexpression virus and selected with puromycin. (A) Polysome profiling was performed using sucrose gradient ultracentrifugation and fractionated into 24 fractions to separate highly translated (more dense) translation complexes. (B) Overlay of representative polysome profiles. Polysome free, early polysome, and late polysome are indicated. (C) Distribution of the *HiBiT-HBG* mRNA across the polysome profile. Dashed lines separate the polysome free, early polysome, and late polysome regions of the profile. (D) Indicated mRNA content in the polysome free, early polysome, and late polysome regions. * $P < .05$. ns, not significant; OE, overexpression.

population, consistent with previous work⁸ (supplemental Figure 3A-C). HbF increased upon *IGF2BP1* overexpression (supplemental Figure 3D). Polysome profiling showed a nonsignificant trend toward *HBG1/2* mRNA enrichment in late polysomes (supplemental Figure 3G).

Given the described function of *IGF2BP1* as an m6A RBP that supports the translation of bound transcripts, we searched for motifs matching that of the *METTL3/METTL14* m6A writer complex motif DRACH in the *HBG1/2* mRNAs. To our surprise, no m6A sites are present in the 3' UTR sequences of *HBG1* or *HBG2* in which functional m6A elements have typically been described across the transcriptome. We observed 10 putative m6A motifs in *HBG1/2*, each located in shared CDS (Figure 6).

We hypothesized that putative m6A motifs in exon 3 CDS could provide selective RNA binding of *HBG1/2*, given that they are stop codon-proximal and absent in *HBB* despite the high degree of homology between these transcripts.^{18,39} Of these, we selected the most stop codon-proximal putative m6A motif and designed *HiBiT-HBG* reporter vectors containing the nonsynonymous mutation c.406A>T (DRACH>DRTCH, p.135Thr>Ser), eliminating the possibility of m6A at this position by replacing the target adenine in the motif, or c.405 G>C (DRACH>DCACH, p.134Val>Val), a synonymous mutation disrupting the second position in the m6A motif.

Given our observation that *IGF2BP1* bound to *HBG1/2* transcripts (Figure 3), we investigated whether the introduction of

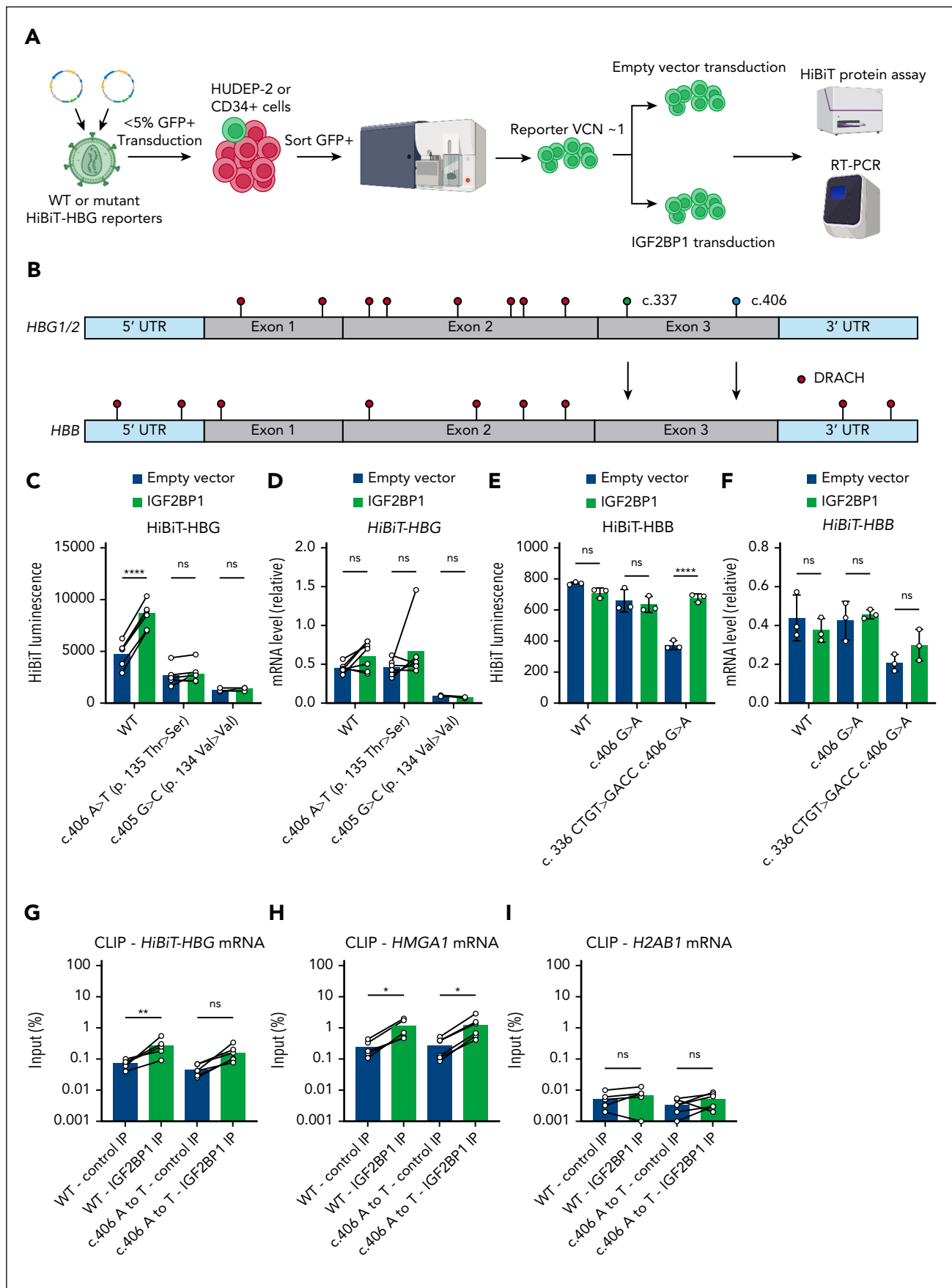


Figure 6.

point mutations at the stop codon–proximal m6A motif could disrupt or mitigate the binding of IGF2BP1 to the *HiBiT-HBG* reporter transgene. CLIP-qPCR in HUDEP-2 cells overexpressing IGF2BP1 with the c.406A>T mutant reporter appeared to demonstrate a reduction in IGF2BP1-bound c.406A>T *HiBiT-HBG* reporter transcript compared to WT reporter transcript; however, this reduction failed to meet the threshold of significance (Figure 6).

To test the impact of these point mutations on gene expression, we transduced HUDEP-2 cells at low multiplicity (<5% transduction frequency) to ensure that different constructs all had comparable vector copy number of ~1. We observed that only the WT *HiBiT-HBG* but not the c.406A>T or c.405G>C mutants could be upregulated by IGF2BP1 overexpression (Figure 6). Notably, the baseline protein level of both mutants was lower than the WT reporter vector. We also tested the mRNA level of the *HiBiT* reporter transcript from time-matched samples upon IGF2BP1 overexpression. We observed that IGF2BP1 failed to increase the transcript level of any of the three reporter constructs consistent with a role for IGF2BP1 in directly supporting the translation of *HBG1/2* through its stop codon–proximal m6A motif.

Addition of HBG-like stop codon–proximal m6A motifs to HBB enables posttranscriptional regulation by IGF2BP1

We observed that the *HBB* transcript lacks a homologous stop codon–proximal m6A motif as the one we mutated in *HBG1/2* to disrupt IGF2BP1-mediated posttranscriptional regulation. Notably, it differs at the paralogous globin position by only a single base-pair. The naturally occurring Calvino *HBB* variant (c.406G>A, p.135Ala>Thr) is a hematologically silent *HBB* variant which creates a “*HBG1/2*-like” m6A motif at this position.⁶¹ We created a WT and c.406G>A version of a *HiBiT-HBB* reporter. However, we observed no change in its mRNA or protein level in response to IGF2BP1 overexpression (Figure 6E–F). Some targets of IGF2BP1 and other m6A reader proteins bind at multiple methylated sites on target mRNA.^{21,39} Notably, *HBG1/2* has 2 putative m6A motifs in exon 3 of its CDS that *HBB* lacks (Figure 6B). We next tested whether addition of a second gamma-like m6A motif to the *HBB* reporter could enable IGF2BP1-mediated posttranscriptional regulation, termed the c.336CTGT>GACC, c.406G>A *HiBiT-HBB* reporter. Upon IGF2BP1 overexpression, the c.336CTGT>GACC, c.406G>A *HiBiT-HBB* reporter, unlike the WT or c.406G>A reporters, exhibited a similar upregulation response to the WT *HiBiT-HBG* reporter at the protein level while the mRNA level remained unchanged (Figure 6E–F). To test whether IGF2BP1 could bind to WT or mutant

HiBiT-HBG reporter transcripts, we performed IGF2BP1 CLIP-qPCR. We observed binding of IGF2BP1 to WT but not c.406 A to T mutant *HiBiT-HBG* reporter transcript. Mutation of this position eliminates the A within the DRACH motif at this position (Figure 6G). Known IGF2BP1 target *HMGA1* was enriched upon IGF2BP1 pulldown in both WT and mutant samples (Figure 6H). *H2AB1* transcript was not recovered in either set of samples (Figure 6I). These results suggest that these 2 stop codon–proximal CDS m6A sites are sufficient to confer IGF2BP1-mediated posttranscriptional regulation.

Given our observation that mutation of key positions with putative m6A motifs could abrogate the direct effect of IGF2BP1 on *HBG1/2* regulation, we investigated m6A presence on the *HBG1/2* mRNA.³⁹ We first performed MeRIP with anti-m6A antibody to test *HBG1/2* mRNA and *HIC2* mRNA enrichment. Because *HBG1/2* and *HIC2* are mostly silenced in adult cells, we performed m6A MeRIP from poly-A⁺ mRNA isolated from HUDEP-2 *BCL11A*^{−/−} cells. We observed significant enrichment of both *HBG1/2* and *HIC2* mRNA by anti-m6A (supplemental Figure 6A). To more directly assess the methylation state of *HBG1/2* mRNA from primary erythroid cells, we performed direct RNA-seq with m6A basecalling (m6A DRS). This method directly assesses m6A modification at single-base-pair resolution without enzymatic processing.^{62–65} To confirm that the m6A basecalling pipeline could accurately assign m6A modification, we generated *HBG2* templates by in vitro transcription without or with m6A-modified (N6 methyladenosine triphosphate) A bases. We confirmed that m6A DRS could accurately identify modified m6A bases at each of the 10 DRACH sites within *HBG2* (supplemental Figure 6B). Next, we purified poly(A) mRNA from cord blood unit reticulocytes (supplemental Figure 6C).⁴⁸ More than 90% of total reads mapped to globin transcripts, consistent with the expected gene expression profile of reticulocytes.⁶⁶ We also performed m6A DRS from expansion phase HUDEP-1 and *BCL11A*^{−/−} HUDEP-2 cells (supplemental Figure 6D). From all these biological samples, we observed modifications at each of the 10 CDS DRACH motifs present in the *HBG1/2* transcripts (supplemental Figure 6E). Together these results indicate that *HBG1/2* transcripts are m6A modified in erythroid precursors.

Discussion

As part of hemoglobin switching, polycomb repressive complex 1 acts to transcriptionally silence at least 3 heterochronic components supporting the fetal gene expression program, *IGF2BP1*, *IGF2BP3*, and *LIN28B*.⁶ In 2017, *IGF2BP1* was first identified as a regulator of HbF, found to be associated with reduced *BCL11A* levels. Several lines of evidence suggest that its repression of *BCL11A* may be indirect, including a lack of

Figure 6. IGF2BP1 promotes expression of HBG1/2 through the stop codon–proximal DRACH motif. (A) Lentivirus carrying the WT or stop codon–proximal DRACH mutant SFFV-*HiBiT-HBG*-PGK-eGFP cassette was used to transduce WT HUDEP-2 cells at low efficiency to ensure low vector copy number to enable comparisons between different variants. GFP⁺ cells were sorted and transduced with either IGF2BP1 or control overexpression vector and selected with puromycin. (B) An alignment schematic of *HBG1/2* and *HBB* mRNAs with DRACH motifs indicated by lollipops. The position of the stop codon–proximal m6A motifs refers to the position of the central “A” in the DRACH motif. (C) *HiBiT-HBG* protein level was measured by *HiBiT* luminescence. WT but not stop codon–proximal mutants were responsive to IGF2BP1 overexpression. (D) *HiBiT-HBG* mRNA levels were measured relative to *glyceraldehyde-3-phosphate dehydrogenase (GAPDH)* in cells at time points matched to panel C. (E) WT, c.406 G>A (p.135 A>T) variant, or c.336 CTGT>GACC (p.112 C>T) c.406 G>A (p.135A>T) *HiBiT-HBB* cells were generated as described in panel A. *HiBiT-HBB* protein level was measured by *HiBiT*-luminescence. (F) *HiBiT-HBB* mRNA levels were measured relative to *GAPDH* in cells at time points matched to panel E. (G–I) HUDEP-2 cells with WT or c.406 A to T mutant *HiBiT-HBG* reporter vector overexpressing IGF2BP1 were used to perform IGF2BP1 CLIP-qPCR. Enrichment of *HiBiT-HBG* mRNA (G), known IGF2BP1 target *HMGA1* (H), and negative control *H2AB1* mRNA (I) is shown. CLIP-qPCR controls are shared with Figure 2, here broken out by WT or c.406 A>T transgene. **P* < .05; ***P* < .01; *****P* < .0001. GFP, green fluorescent protein; IP, immunoprecipitation; ns, not significant; VCN, vector copy number.

change in *BCL11A* translation or RNA stability upon *IGF2BP1* overexpression, and the typical role of *IGF2BP1* in supporting the expression of its targets.^{8,15} Our work demonstrates that *IGF2BP1* acts through a bipartite mechanism to regulate HbF expression. We show that the component of this regulation dependent upon *BCL11A* occurs via control of *HIC2*. Recent work has shown that *HIC2* is a target of *let-7* miRNA, and that *HIC2* acts as a direct transcriptional repressor of *BCL11A*.^{4,56} *IGF2BP1* has been shown to inhibit miRNA-mediated inhibition of mRNA targets.⁵⁵ Our work identifies a role for *IGF2BP1* in supporting the expression of *HIC2*, thus indirectly repressing *BCL11A* transcription. Of the 115 DRACH sites on the *HIC2* mRNA, 91 are located in the 3' UTR. This, in conjunction with the fact that *HIC2* mRNA is enriched by m6A MeRIP and both *IGF2BP1* and *HIC2* are predominantly expressed in the fetal stage, is consistent with a mechanism through which *IGF2BP1* protein directly binds the 3' UTR of *HIC2* mRNA to support its expression.

When *IGF2BP1* is introduced to adult erythroid cells in which it is typically silenced, this leads to reactivation of HbF expression. Hemoglobin switching is understood to be driven by expression of strong transcriptional repressors including *ZBTB7A* and *BCL11A*.^{1,29} We have shown that reintroduction of the fetal factor *IGF2BP1* to adult cells acts through at least 2 mechanisms to activate HbF expression, *BCL11A*-dependent transcriptional regulation and *BCL11A*-independent posttranscriptional regulation. Notably, the transcript level determined by transcriptional regulation is likely to influence the effect size of any posttranscriptional regulation. We observe that the absolute difference in HbF level is greater in *BCL11A*^{-/-} than *BCL11A*^{+/+} HUDEP-2 cells (Figure 1C), suggesting that the contribution of transcriptional control may predominate.

The extent of HbF reactivation observed in this work upon overexpression of *IGF2BP1* is of lesser magnitude as compared with previous published work. In this study we used a spleen focus-forming virus (SFFV) promoter-driven construct to drive *IGF2BP1* overexpression as compared with the erythroid-specific *SPTA1* promoter used in this previous work.⁸ We observed ~1e4 copies per nanogram of complementary DNA from HUDEP-2 cells overexpressing *IGF2BP1* mRNA with our SFFV-driven system (supplemental Figure 3E) as compared with ~1.5e5 copies per nanogram complementary DNA in the previous work, suggesting that a lesser degree of *IGF2BP1* overexpression may account for the difference in HbF expression effect size. It is possible that erythroid-specific *SPTA1* (spectrin alpha, erythrocytic 1) promoter could drive *IGF2BP1* overexpression during terminal erythroid maturation, perhaps potentiating posttranscriptional effects. We cannot exclude that postribosome loading-mediated translational regulation of *BCL11A* could be a contributing factor, especially at high levels of *IGF2BP1* overexpression.

The posttranscriptional *BCL11A*-independent mechanism of *IGF2BP1* acting directly on *HBG1/2* mRNA identified in this work is supported by observations of *IGF2BP1* binding, translation regulation, and the necessity and sufficiency of stop codon-proximal m6A motifs. Regarding binding, we observe a lower proportion of bound transcript than the positive control transcript *HMGA1* and a higher proportion than the negative control *H2AB1* (Figure 3). *IGF2BP* family members are known

to bind in the *HMGA1* 3' UTR.⁶⁷ Given that m6A motifs are present only in the *HBG1/2* CDS, we speculate that actively translated *HBG1/2* mRNA would be unbound by *IGF2BP1*, as this protein would be sterically removed as part of the pioneer round of translation.⁶⁸ Although we did not observe a significant increase in the level of transcript using our HiBiT-HBG reporter upon *IGF2BP1* overexpression (Figure 6D), we do see such an increase in the *HBG1/2* RNA level in *BCL11A*^{-/-} HUDEP-2 cells (Figure 1B). Depending on the context of the cell state, expression level, methylation, and perhaps other RNA-binding factors, *IGF2BP1* may preferentially regulate *HBG1/2* translation and/or RNA stability.

Notably, *HBB* lacks the m6A motifs present in *HBG1/2*. Although m6A modification has become appreciated as a dynamic regulator of gene expression, most well-understood examples have been characterized within the 3' UTR.^{36,53,69} Our report adds to an emerging body of work pointing toward CDS m6A as a regulatory feature, in this instance a mechanism supporting the expression of *HBG1/2* mRNA driven by m6A marks in the CDS bound by the m6A reader *IGF2BP1*. Several lines of evidence support the importance of CDS m6A, particularly in the vicinity of the stop codon. First, the conserved nature of some marks; second, several well-characterized examples from RNA biology before the explosion of work on m6A in the 2010s (particularly *c-MYC*, one of the first well-characterized targets of *IGF2BP1*); and third, recent studies characterizing a novel developmentally associated RNA decay mechanism termed CDS-m6A decay, which requires the m6A reader YTHDF2.^{18,20,22,59,60,70}

As part of investigating the direct effect of *IGF2BP1* on *HBG1/2* mRNA, we investigated whether these transcripts were m6A modified through 2 orthogonal approaches. Although some in vitro evidence suggests high affinity (~15nM to the β -actin zipcode binding protein 1 element) binding of the *IGF2BP1* KH34 domain (responsible for RNA binding) to unmodified target RNA, most functional studies of *IGF2BP1* have assumed m6A methylation is required for *IGF2BP1*-mediated regulation of its bound targets.^{37,39,71} We observed enrichment of *HBG1/2* transcripts from HUDEP-2 *BCL11A*^{-/-} by m6A MeRIP (supplemental Figure 6A). Of note, *HBG1* and *HBG2* lack predicted DRACH motifs in their UTRs. Based on DRS we observed m6A modifications at every *HBG1/2* CDS DRACH motif (supplemental Figure 6E).

We speculate that *IGF2BP1* binding at CDS m6A modifications may help to suppress CDS-m6A decay by interfering with YTHDF2 binding to the developmentally regulated *HBG1/2* mRNA, a protective measure unneeded by *HBB*, which lacks stop codon-proximal m6A sites in the CDS that may promote translational pausing. This would be consistent with a similar mechanism proposed for *IGF2BP1* target *c-MYC* before the current understanding of m6A as a dynamic regulatory mark, specifically, that *IGF2BP1* binding (also known as CRDBP) may bind to the ~250 nucleotide coding CRD, would protect a downstream endonucleolytic cleavage site within the CRD that could become exposed by ribosomal pausing (thought, at that time, to be due to rare codon usage within the element).^{20,21}

In conclusion, we have identified that the reintroduction of *IGF2BP1* in adult erythroid cells supports the expression of

the developmentally regulated *HBG1/2* genes through a bipartite mechanism. First, there is an indirect transcriptional regulation whereby *IGF2BP1* supports *HIC2* expression, directly enabling the transcriptional silencing of *BCL11A*. Notably, *HIC2* mRNA is a target of let-7, and *IGF2BP1* has been reported to have a role in inhibiting miRNA-mediated regulation by binding near miRNA binding sites in the 3' UTR of transcripts.^{55,56} Second, *IGF2BP1* directly binds to *HBG1/2* transcripts at stop codon-proximal m6A motifs, and this binding supports their translation. Introducing *HBG*-like m6A motifs to the *HBB* transcript boosts its *IGF2BP1*-dependent expression. In addition to our observation of *IGF2BP1*-mediated direct posttranscriptional support of *HBG1/2* expression, recent work has identified the RBP *PUM1*, a *KLF1* target gene, as a negative posttranscriptional regulator of *HBG1*.⁷² These results suggest a previously underappreciated layer of hemoglobin switching regulation, that is, a physical relationship between heterochronic RBPs and developmentally regulated globin genes.

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Authorship

Contribution: S.C. and D.E.B. conceptualized the study, and wrote, reviewed, and edited the manuscript; S.C., G.-H.H., and D.E.B. developed the methodology; S.C., T.W., G.-H.H., M.H., D.V., M.S., J.Z., F.A., A.G., A.S., and D.E.B. performed investigations; S.C. prepared the original manuscript draft; D.E.B. acquired funding; and A.S. and D.E.B. supervised the study.

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Footnotes

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Other data and protocols are available from the corresponding author, Daniel E. Bauer (bauer@bloodgroup.tch.harvard.edu), on request.

The online version of this article contains a data supplement.

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